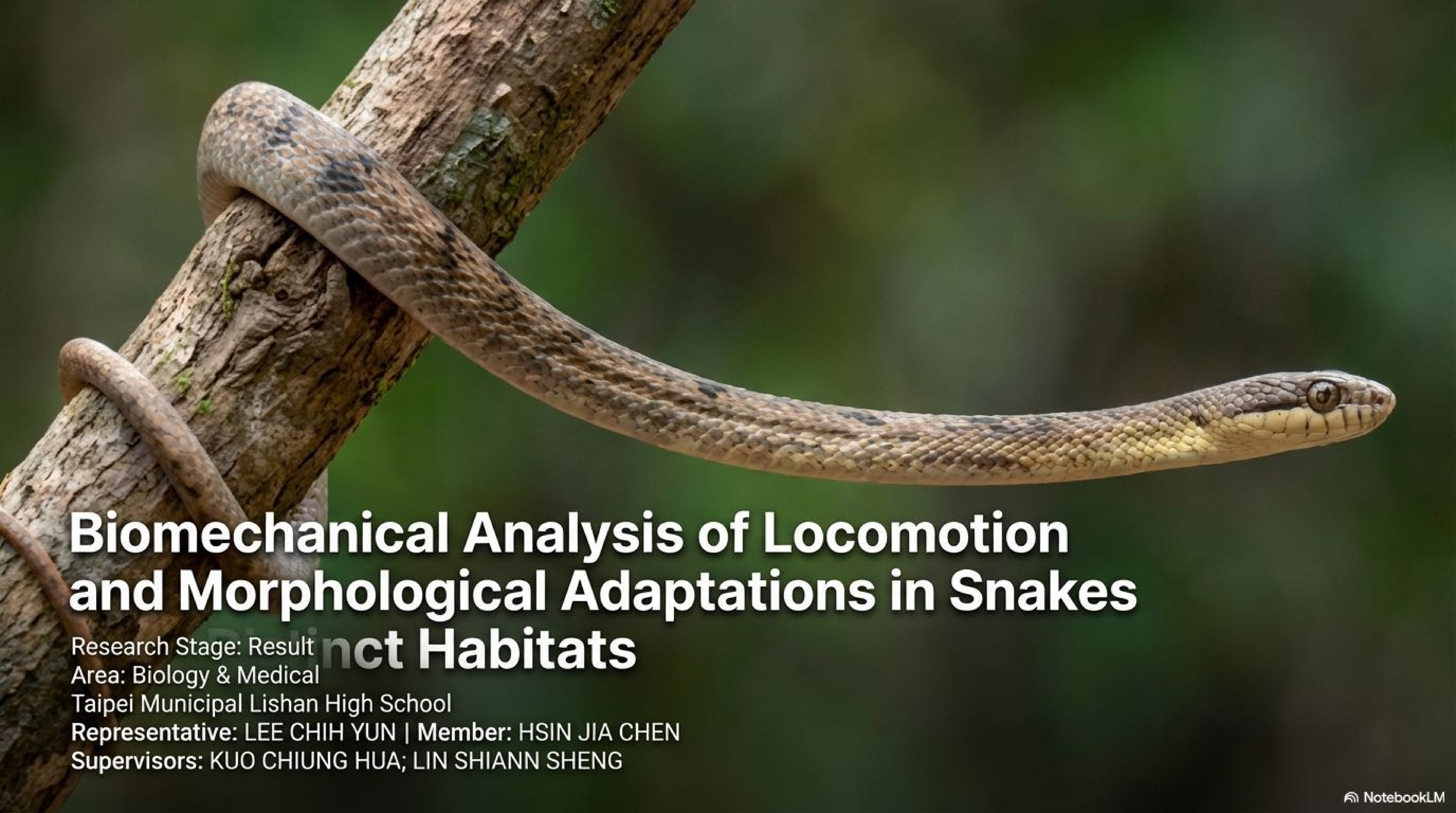

Biomechanical Analysis of Locomotion and Morphological Adaptations in Snakes Distinct Habitats

Research Stage: Result
Area: Biology & Medical
Taipei Municipal Lishan High School
Representative: LEE CHIH YUN | **Member:** HSIN JIA CHEN
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The Spark: A Campus Encounter with Gravity-Defying Mechanics

Observation:

During a campus observation, we encountered a wild Boiga kraepelini. We were surprised to find that this species, within the forest canopy, can coil their tails onto our fingers and extend to survey its surroundings.

The Question:

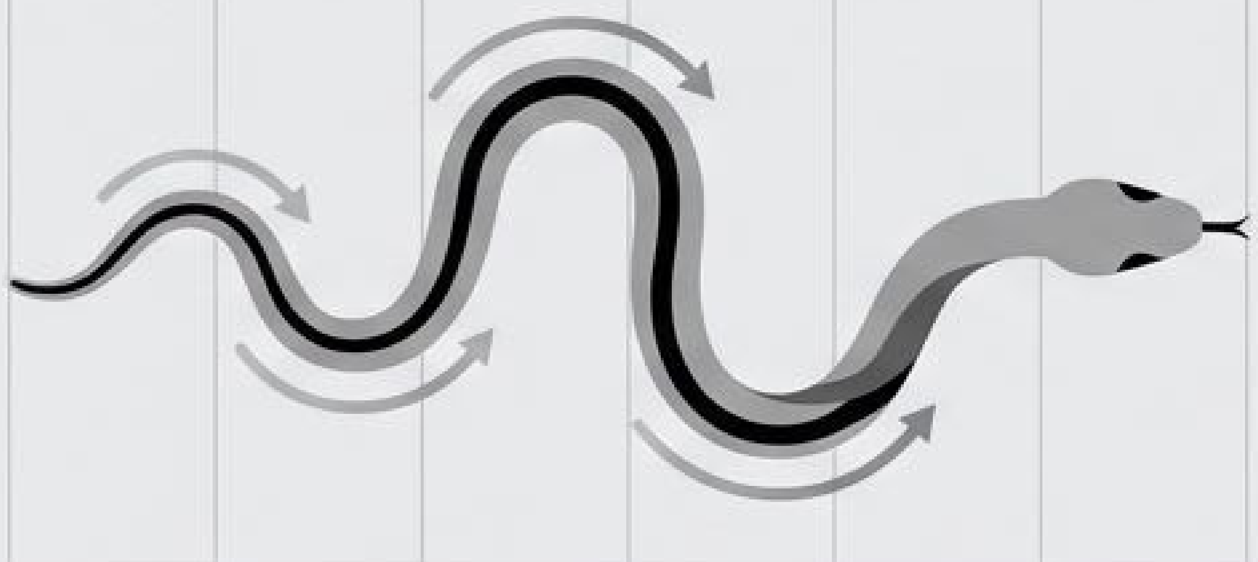
This specific observation prompted our investigation: What specific anatomical structures grant them an advantage in climbing?

Key Concept:

The '**Cantilever**' ability—supporting weight on one end while extending into empty space.

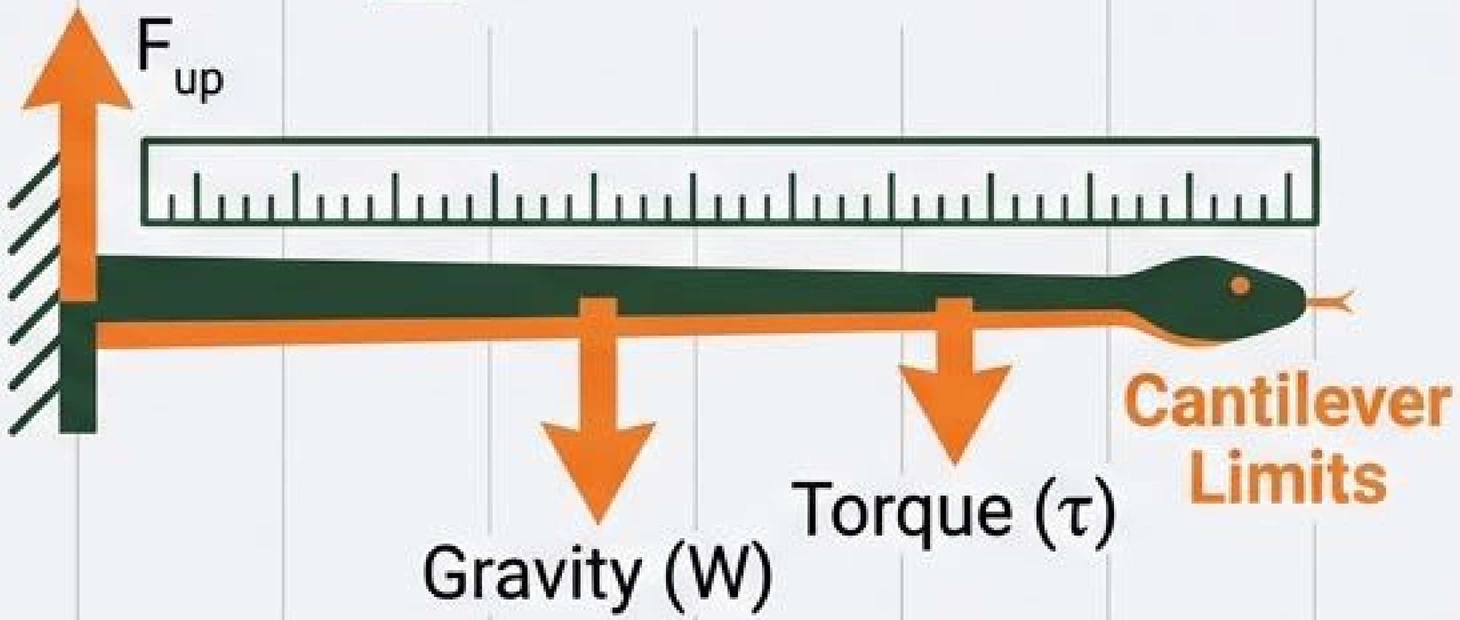
Moving Beyond Qualitative Patterns to Quantitative Limits

QUALITATIVE



Current Literature: Focuses on crawling patterns.

QUANTITATIVE



This Study: Focuses on physical failure points and Cantilever Limits.

The Gap: There is a lack of quantitative data regarding the specific biomechanical limits of snakes when challenging gravity.

Novelty: We aim to correlate these physical limits with specific morphological adaptations to quantify how form defines function.

Experimental Subjects: A Spectrum of Ecological Niches

Boiga kraepelini

Arboreal.
Length: 86cm.
Habitat:
Taiwan/China.

Heterodon nasicus (Western Hognose)

Terrestrial/Fossorial.
Length: 47cm.

Pantherophis guttatus (Corn Snake)

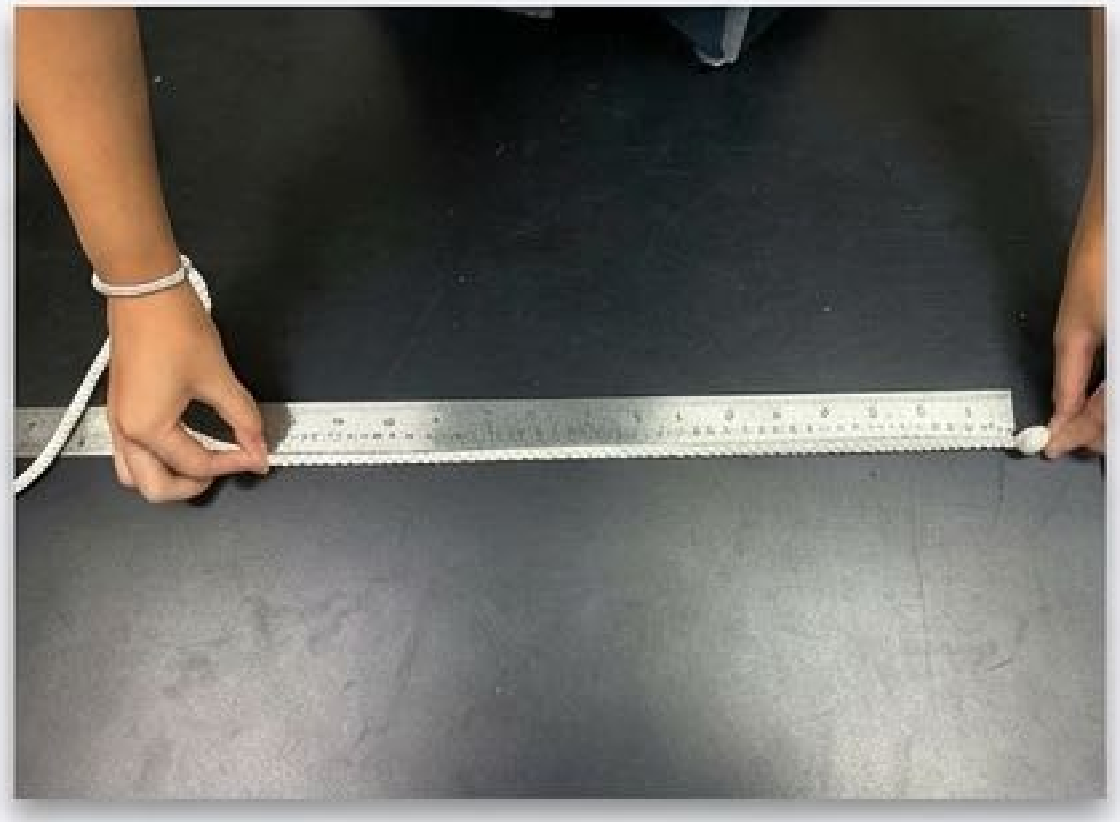
Semi-Arboreal.
Length: 97cm.

Boa constrictor

Mixed Habitat.
Length: 150cm.

Reference Scale Comparison.

Methodology I: Quantifying Morphological Proportions



Length Protocol

Align snake with knotted rope, mark head/tail tips, measure rope to minimize stress.

Segmentation

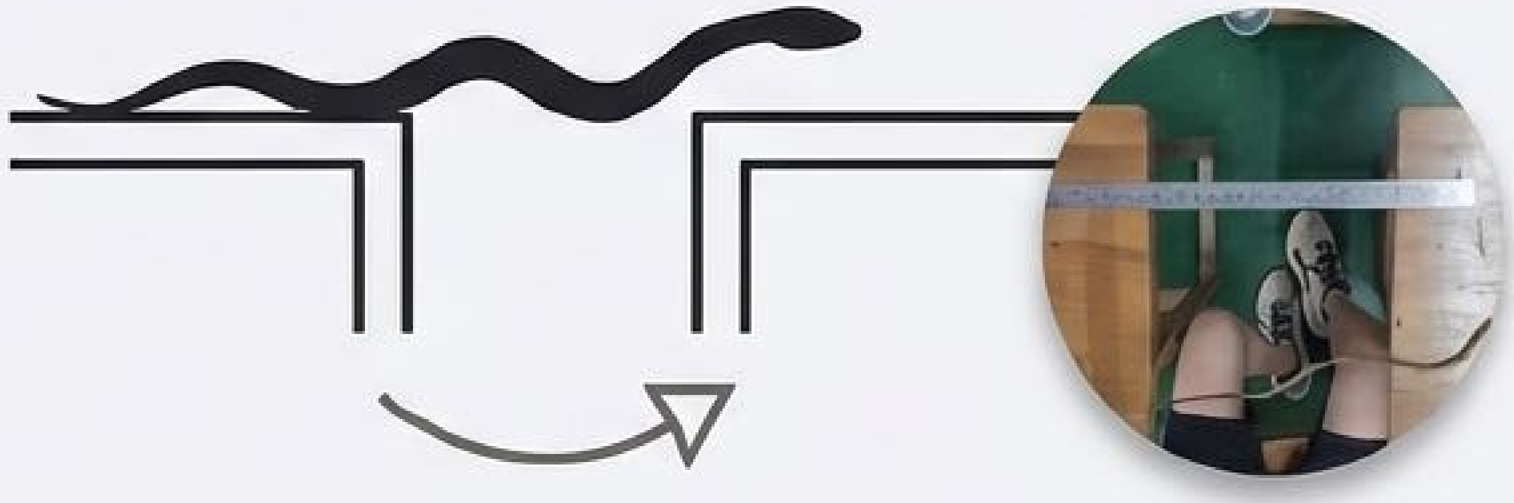
Defined landmarks at (a) posterior margin of head, (b) lowest neck point, (c) base of tail (cloaca).

Width & Rigor

Digital caliper measurements at widest points. All measurements repeated 3 times and averaged.

Methodology II: Stress Testing the Biomechanical Limits

Horizontal Reaching



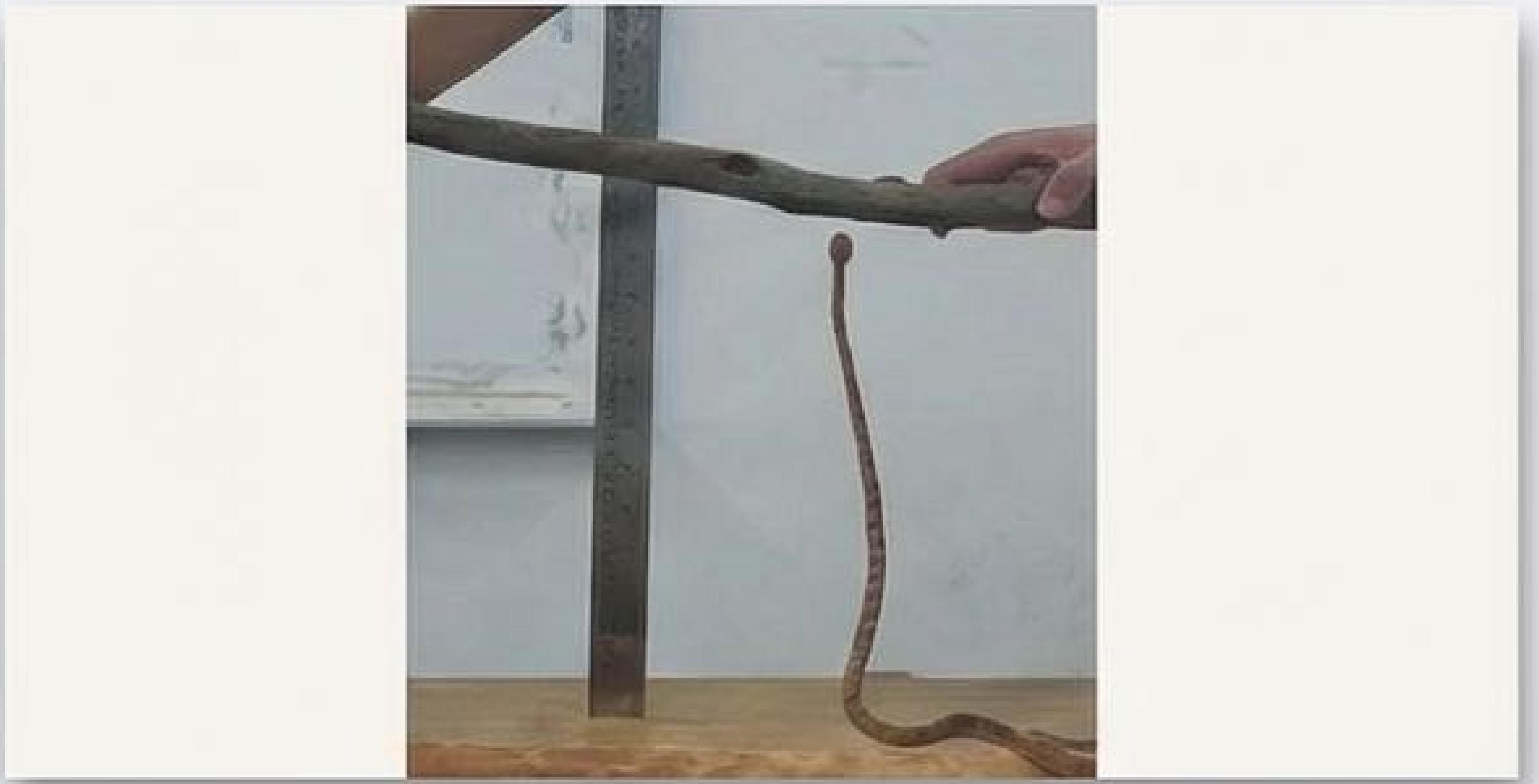
Horizontal Reaching Test.
Goal: Test cantilever limit in absence of support.
Procedure: Distance increased systematically until failure.

Vertical Extension

Vertical Extension Test
Goal: Test ability to overcome gravity.
Metric: Climbing out within 5 minutes.
Procedure: Height increased until failure (0-1 successful trials).

Experimental Verification and Protocol

Horizontal Reaching Setup



Vertical Climbing Setup

Design Evolution

Switched to standardized containers (vertical) and desks (horizontal) to ensure precise height differentials and reduce interference.

Execution

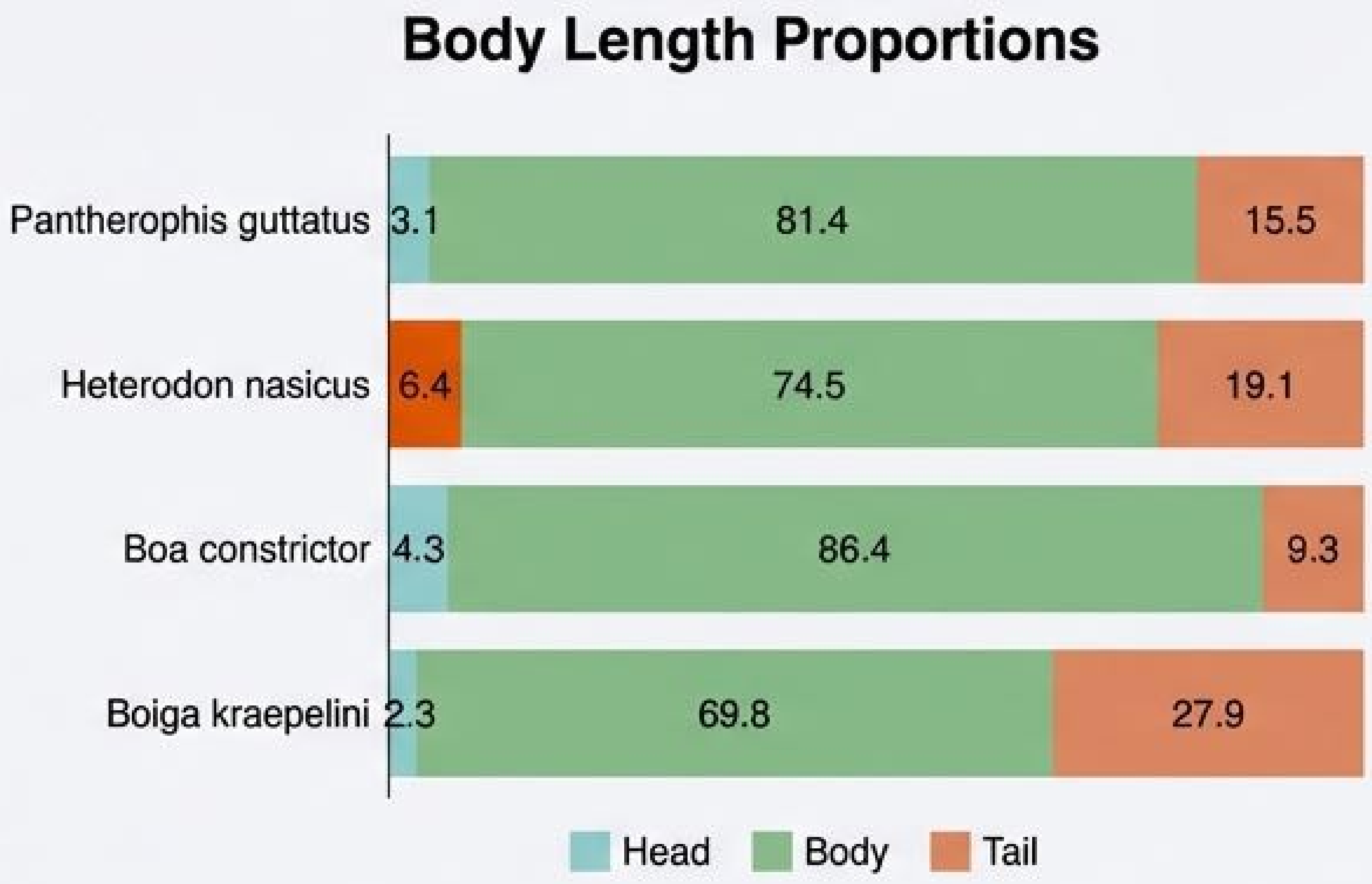
3 trials per increment. Video analysis used to observe muscle control.

Safety & Ethics

Strict adherence to safe husbandry and SDG 15 guidelines. Maintained safe distance to ensure snake willingness.

Results I: Morphological Divergence

Species	Head (%)	Body (%)	Tail (%)
Pantherophis guttatus	3.1	81.4	15.5
Heterodon nasicus	6.4	74.5	19.1
Boa constrictor	4.3	86.4	9.3
Boiga kraepelini	2.3	69.8	27.9



Boiga kraepelini exhibits a massive **27.9%** tail ratio (Posterior Anchor), while *Heterodon nasicus* has a heavy **6.4%** head ratio (Anterior Torque).

Results II: The Horizontal Reaching Limit

Heterodon nasicus (47cm)

Max Reach: 31.9%

Success dropped sharply beyond 15cm. Heavy head = poor stability.

Boiga kraepelini (86cm)

Max Reach: 38%

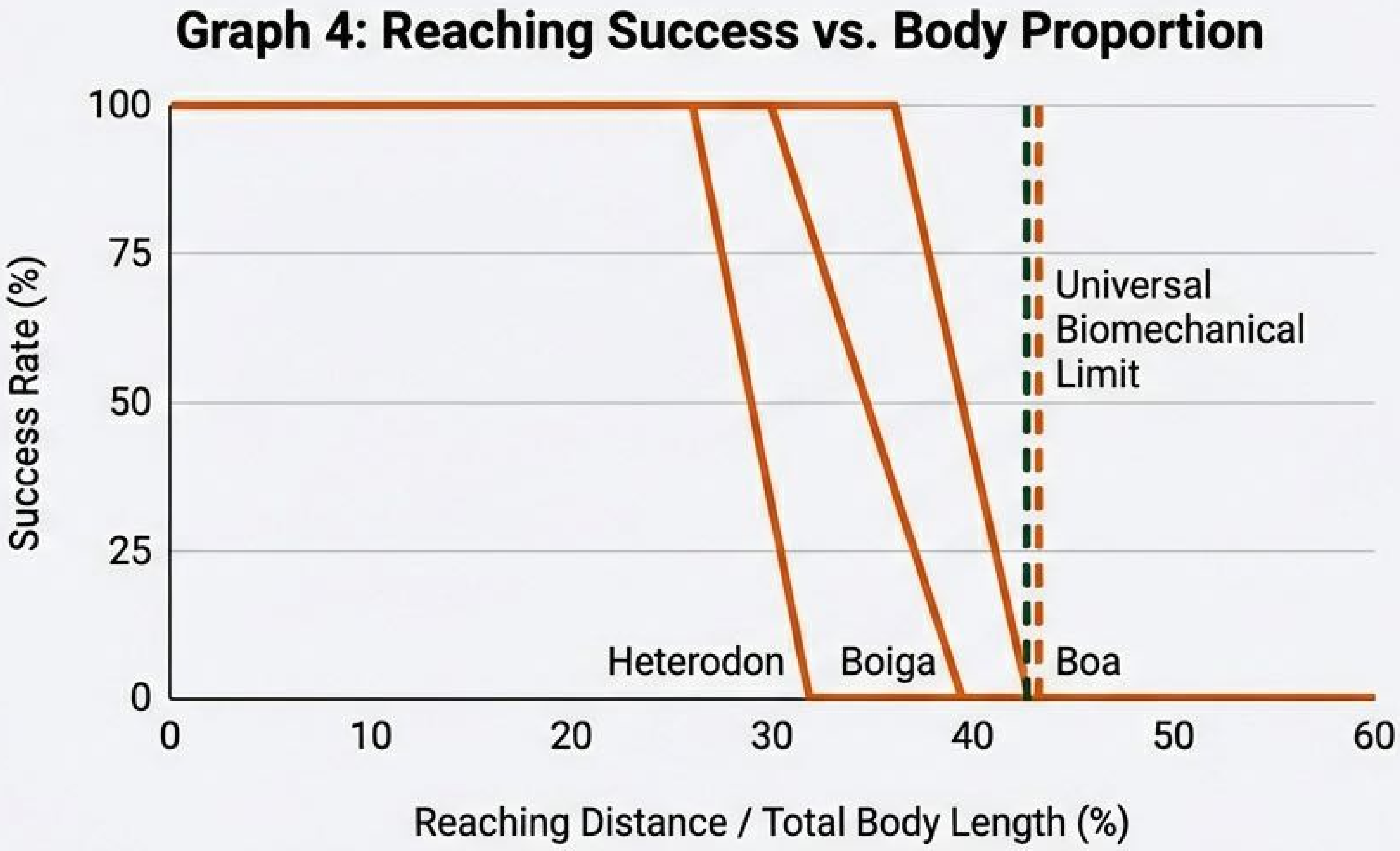
Lower than expected. Thin body resulted in weaker friction on flat surfaces.

Boa constrictor (150cm)

Max Reach: 43%

Highest reach. Massive muscle mass sustains cantilever despite small tail.

Results III: The Universal ~43% Biomechanical Constraint



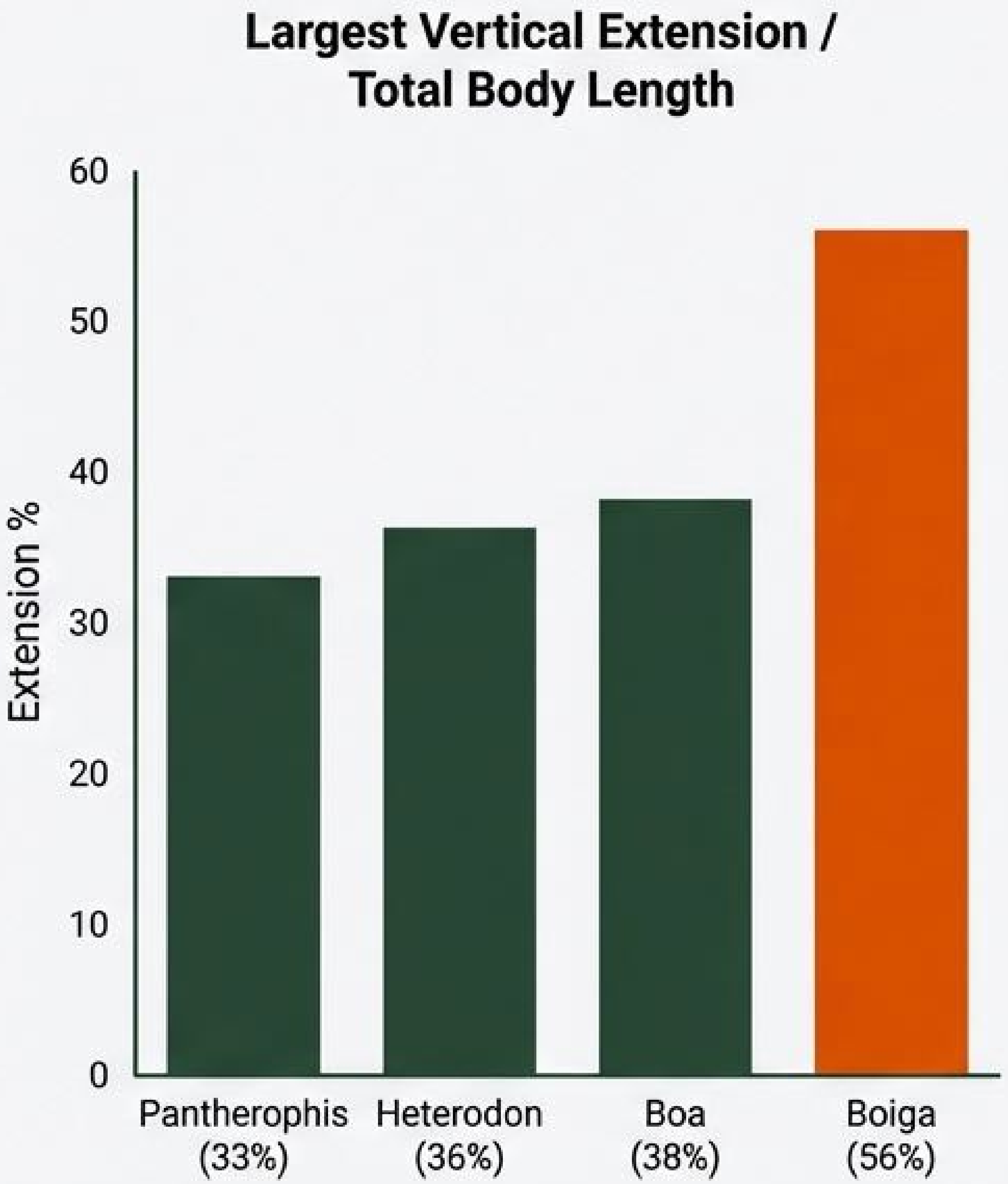
Convergence: Despite size differences (47-150cm), the limit converges around **43%**.

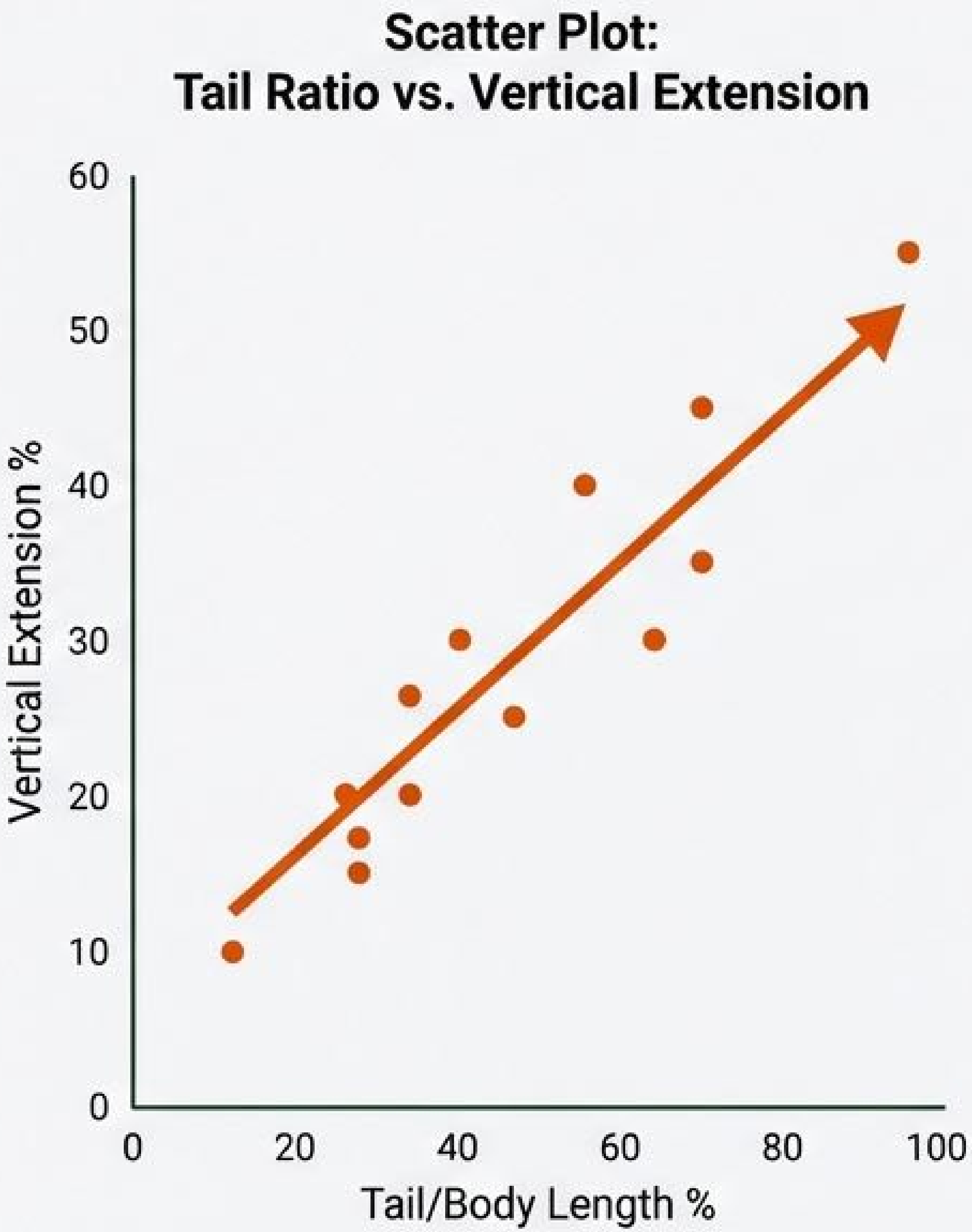
Physics: The point where gravitational torque exceeds muscular counter-torque.

Results IV: Vertical Extension Performance

Boiga kraepelini	Dominant. Max record: **55.8% (48cm)**. Average: **48.3%**
Heterodon nasicus	Poor. Vertical limit approx: **32-42%**. Large head caused balance loss.
Pantherophis guttatus	Moderate. Success rate: **100% at 20cm**, dropping linearly to **33% at 31cm**.
Boa constrictor	**Variable**. Success rate: **0% at 20cm** in smooth container; **100% at 57cm** in high-friction box.

Results V: The Tail as a Posterior Anchor





- **Correlation:** Snakes with **larger tail ratios** perform better against gravity.
- **Mechanism:** The tail acts as a **counterweight and friction anchor** (Posterior Torque Compensation).

Case Study: The Fossorial Trade-off (*Heterodon nasicus*)

Large Head Ratio
(6.4%)

Performance:

Success dropped significantly at just 31.9% extension.

Interpretation:

Form defines function. A heavy head is specialized for digging (fossorial lifestyle) but increases Anterior Torque, making it the least stable species against gravity.

Strategies to Beat Gravity: Rigidity vs. Flexibility

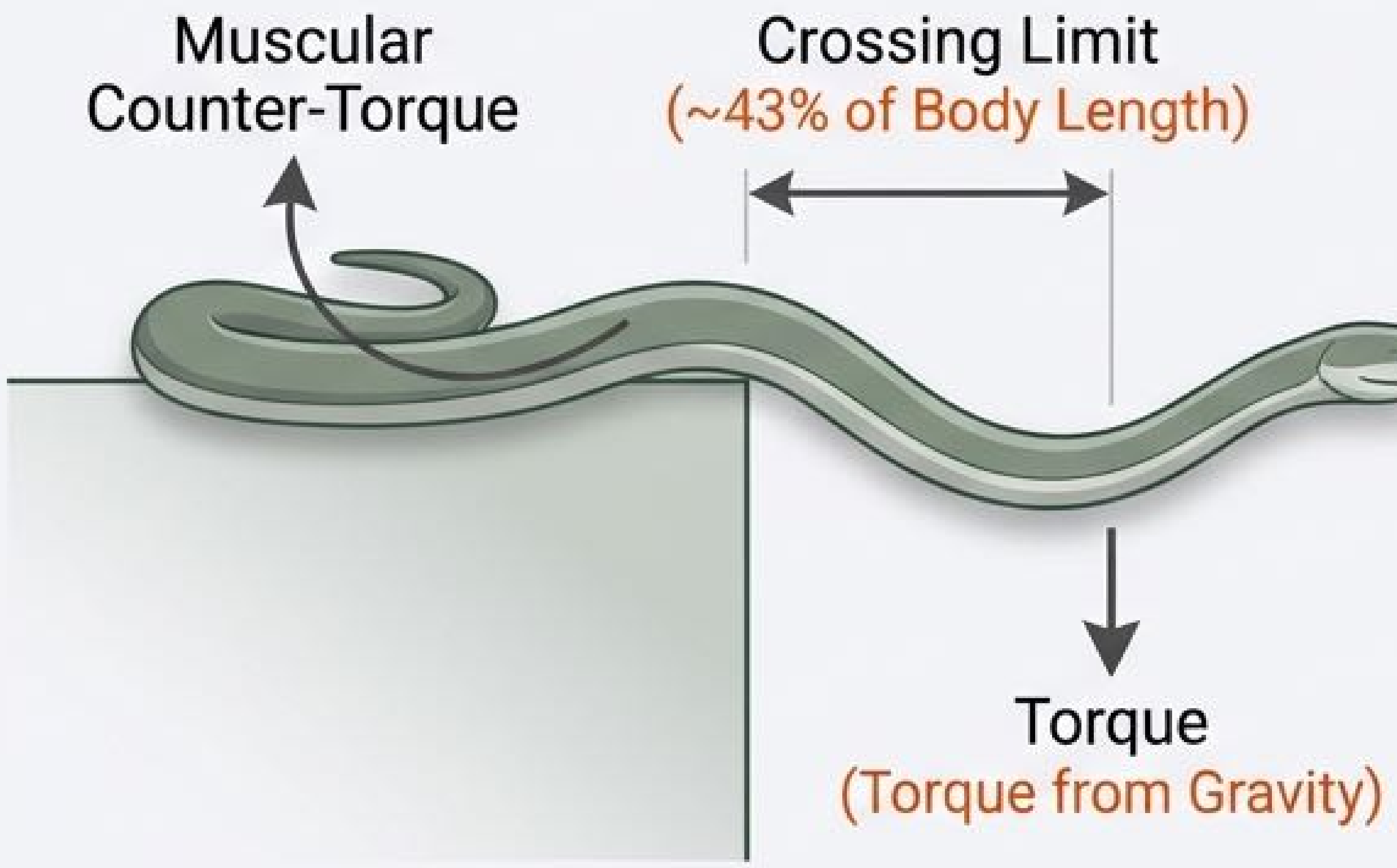
Strategy A: Raw Power.

Short tail (**9.3%**) but high body ratio (**86.4%**). Relies on Rigidity and core muscle strength. Dependent on friction.

Strategy B: Lever Mechanics.

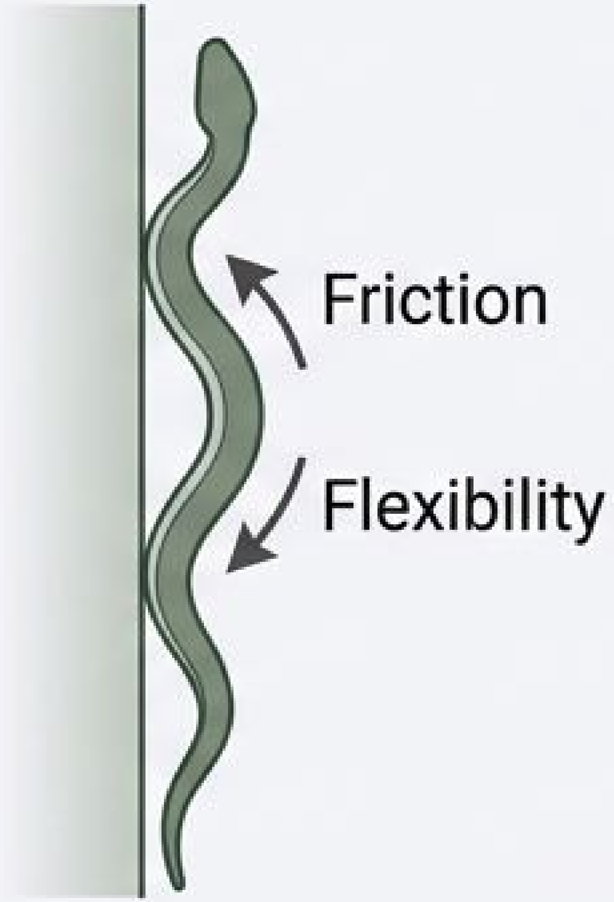
Long tail (**27.9%**) and slender body. Relies on Flexibility and the tail as a counterweight. Superior vertical agility.

The Biomechanics of Failure

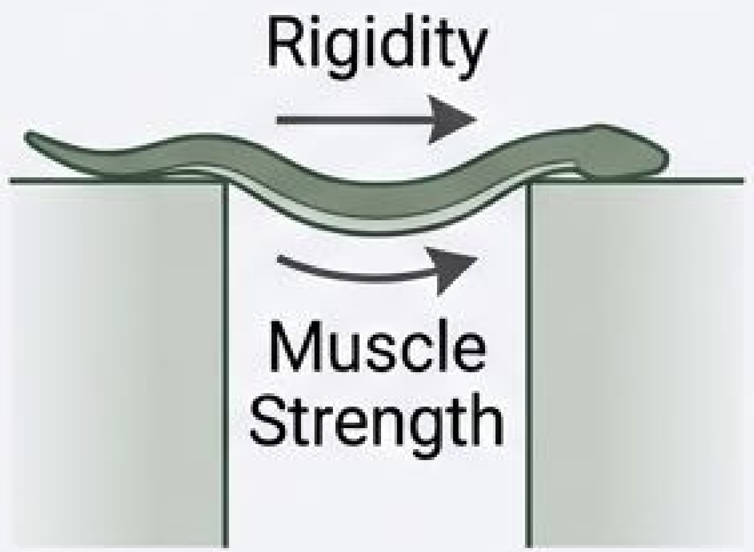


Physical Constraints on Lateral Movement: The crossing limit of approximately **43%** of body length likely represents a widespread **biomechanical bottleneck** in snake body structures. When the gravitational torque on the suspended portion exceeds the counter-torque generated by core muscles, the structure fails.

Strategic Divergence in Vertical Extension and Horizontal Reaching:

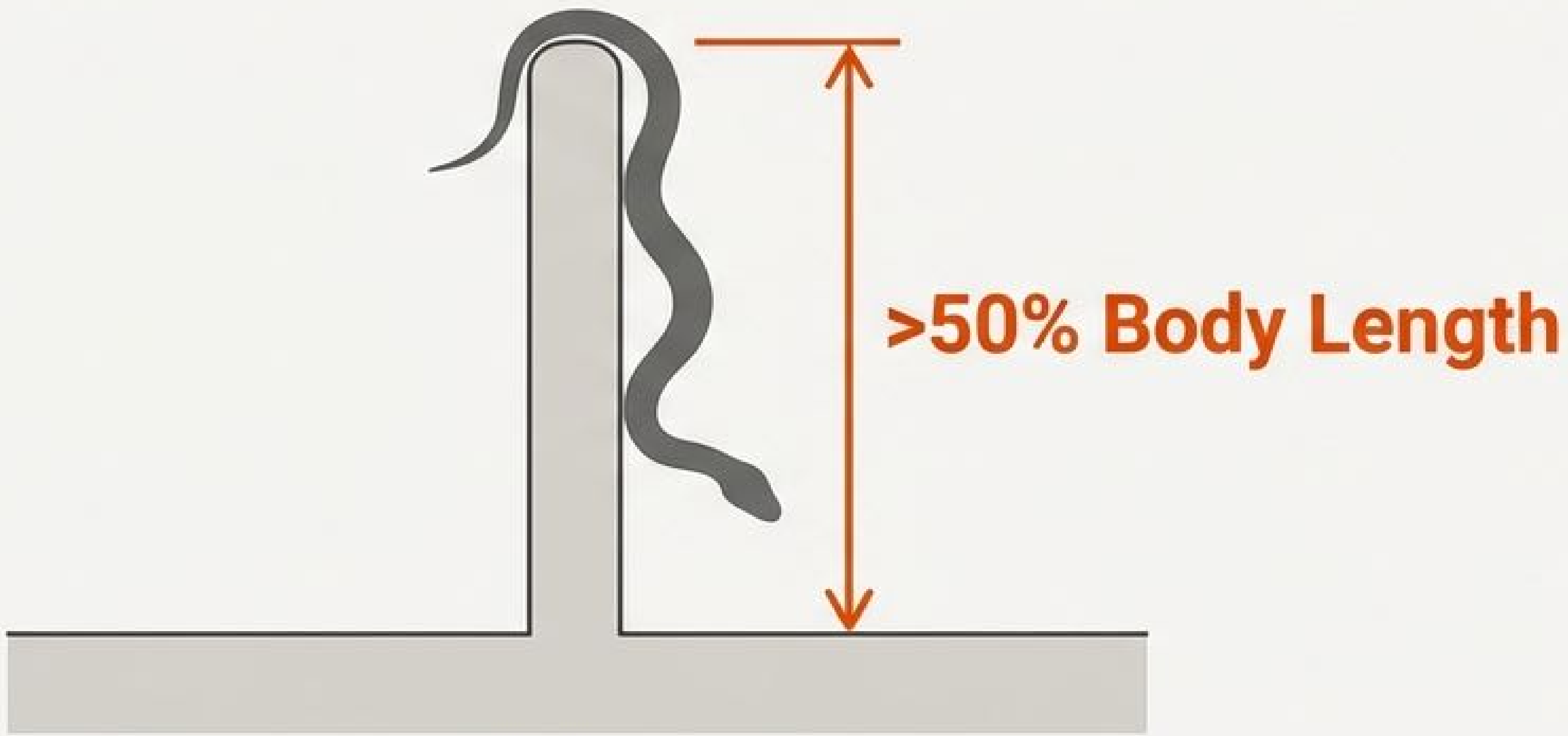


Vertical favors friction/flexibility; favoring slender, lightweight snakes.



Horizontal favors rigidity/muscle; favoring muscular body types.

The 50% Rule



The Definitive Finding.

A snake cannot vertically scale a smooth barrier that exceeds 50% of its total body length without additional footholds.

Significance: A quantitative metric for predicting snake mobility based solely on body length.

Application I: Scientifically Informed Coexistence

15 LIFE ON LAND

Designing Barriers:

Using the 50% Rule to design walls that keep snakes out of residential areas without harm.

Designing Corridors:

Arboreal species are vulnerable on flat ground. We propose "Canopy Bridges" to reduce roadkill, directly supporting SDG 15.

Application II: The Tail as a 'Fifth Limb' for Robotics

Challenge:

Current snake robots struggle with **gap crossing**.

Biomimetic Solution:

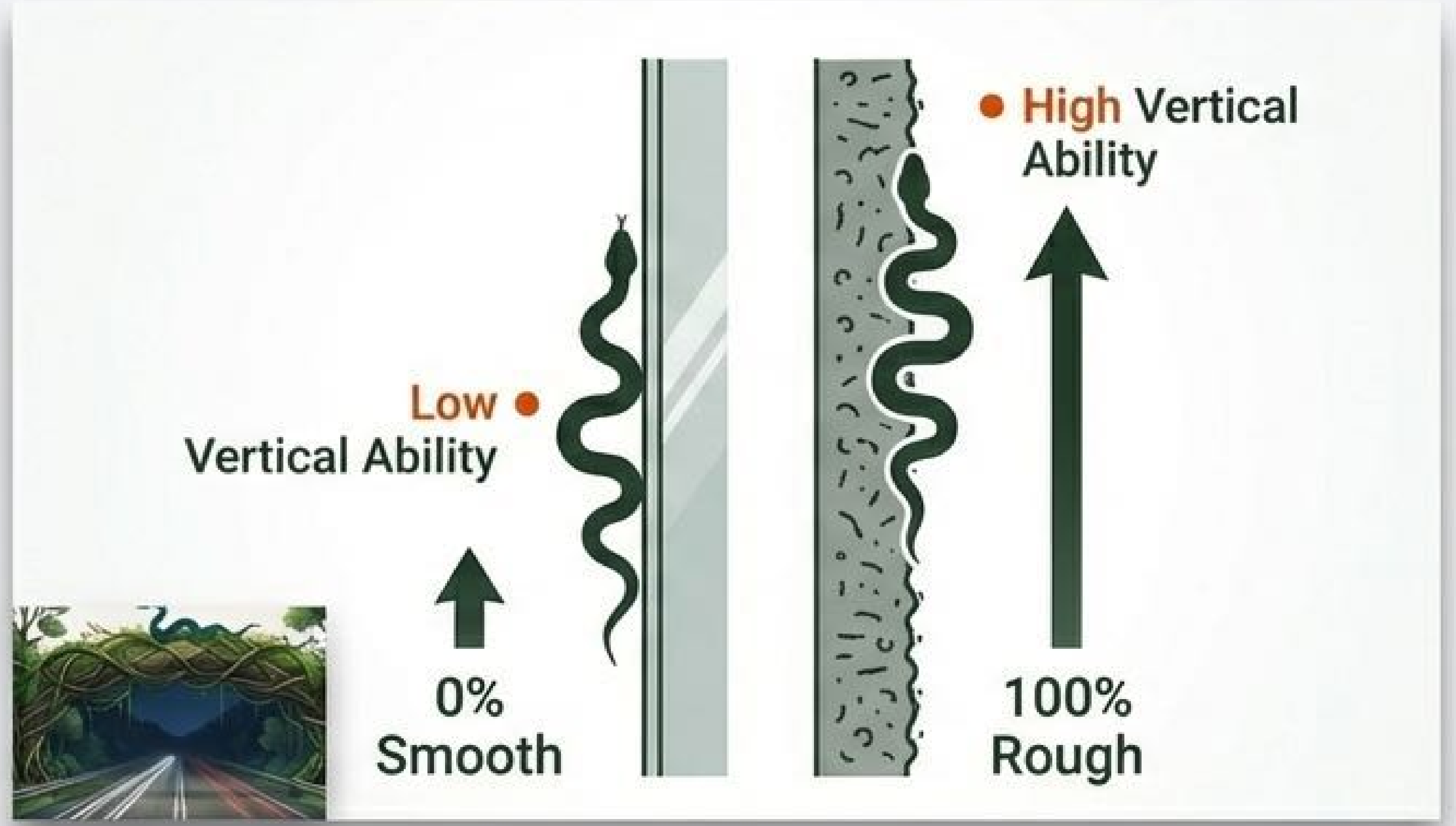
Mimic the **Boiga strategy**.

Proposal:

Engineers should design robots with a **high-tail-ratio structure** to act as a **posterior anchor** and counterbalance, enabling stable **cantilevered extensions** in disaster zones.

Future Potential and Research Limitations

The Friction Variable:



Our Boa tests showed vertical ability is highly dependent on surface friction (0% smooth vs 100% rough).

Next Steps:

1. Expand dataset to more species to verify the "**43% constant**".

2. Collaborate with biophysicists to study how scale micro-structures influence friction during vertical climbing.

References & Acknowledgments

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2. Byrnes, G., & Jayne, B. C. (2012). The effects of three-dimensional gap orientation on bridging performance.
3. Jayne, B. C., et al. (2015). Why arboreal snakes should not be cylindrical.

By understanding the biomechanical limits of nature,
we unlock new possibilities for engineering and coexistence.

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